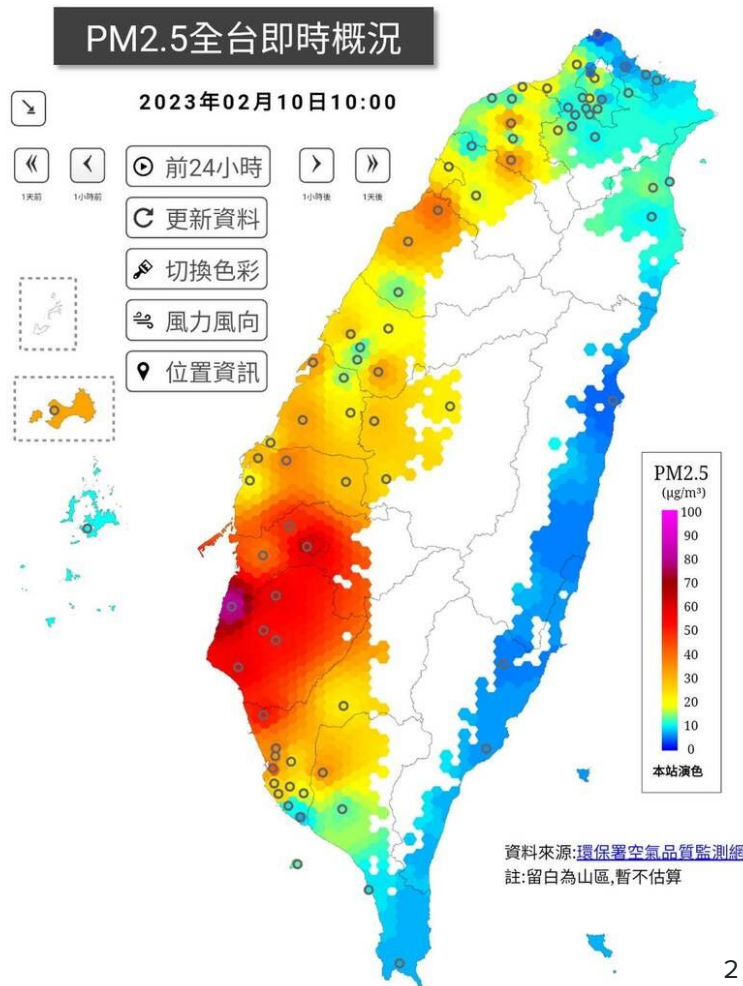


2023 Data Mining

HW1

Task introduction

- PM2.5 prediction
 - Implement linear regression **using only numpy** to predict the value of PM2.5
- Requirement
 - Upload your submission to Kaggle
 - Submit a report and your source code to E3
- Deadline is 4/11 23:59, no late submission



Dataset

2021 Hsinchu meteorological observation data form Central Weather Bureau.

- train.csv
 - Climate data for the first 20 days of each month.
 - [link](#)
- test_X.csv
 - Sample the continuous 10 hours data from the remaining 10 days of each month,take the data of the previous 9 hours as a feature, and the last hour as the answer, get total 244 unique test datas.
Please use the features to predict PM2.5.
 - [link](#)

Training Data

Location	Date	ItemName	0	1	2	3	4	5	6	7	8	9
Hsinchu	2021/1/1 00:00	AMB_TEM	11.1	11.2	11.4	11.5	11.6	11.7	11.9	12.1	12.7	13.9
Hsinchu	2021/1/1 00:00	CH4	2.01	1.99	2	2.02	2.03	2.02	2.02	2.01	2.03	2.03
Hsinchu	2021/1/1 00:00	CO	0.31	0.28	0.28	0.33	0.32	0.26	0.25	0.27	0.32	0.34
Hsinchu	2021/1/1 00:00	NMHC	0.1	0.1	0.08	0.09	0.1	0.07	0.07	0.08	0.1	0.09
Hsinchu	2021/1/1 00:00	NO	1.5	1.4	1.4	1.5	1.4	1.3	1.4	1.9	3.2	5.1
Hsinchu	2021/1/1 00:00	NO2	11.9	10.4	9.8	12.1	12.4	9.2	8.5	9.3	12.4	13.2
Hsinchu	2021/1/1 00:00	NOx	13.5	11.9	11.2	13.7	13.9	10.6	10	11.2	15.6	18.4
Hsinchu	2021/1/1 00:00	O3	21.6	25.1	25.6	22.4	21.1	26.5	25.4	25.3	22.9	24.4
Hsinchu	2021/1/1 00:00	PM10	38	29	27	24	29	22	26	26	31	32
Hsinchu	2021/1/1 00:00	PM2.5	25	24	13	14	15	12	10	10	11	11
Hsinchu	2021/1/1 00:00	RAINFAI	0	0	0	0	0	0	0	0	0	0
Hsinchu	2021/1/1 00:00	RH	64	65	63	63	63	63	63	63	62	57
Hsinchu	2021/1/1 00:00	SO2	#	2.1	2.1	1.8	1.1	0.7	0.8	0.8	#	1.6

#,*,x,A, represented as an invalid value

Testing Data

index_0	AMB_TE	18.2	17.8	17.5	17.5	17.7	18.1	18.2	18.7	20.3	
index_0	CH4	2.41	2.61	2.65	2.87	2.25	2.24	2.45	2.59	2.24	
index_0	CO	0.77	0.74	0.63	0.6	0.36	0.31	0.48	1.01	1.05	
index_0	NMHC	0.29	0.34	0.34	0.37	0.18	0.15	0.24	0.43	0.35	
index_0	NO	6.8	11.1	9.6	13.6	3.1	2.4	17.8	49.5	41.1	
index_0	NO2	30.9	28.2	25.9	22.8	16.5	15.8	21.3	25	26.1	
index_0	NOx	37.7	39.3	35.6	36.4	19.6	18.3	39.1	74.5	67.2	
index_0	O3	4.1	2	1.9	1.8	7.4	6.2	2.2	3	6.3	
index_0	PM10	53	50	36	39	23	21	22	25	36	
index_0	PM2.5	35	35	24	28	15	11	14	17	17	
index_0	RAINFAI	0	0	0	0	0	0	0	0	0	
index_0	RH	84	85	85	85	81	77	77	76	69	
index_0	SO2	2.8	1.9	1.9	1.9	1	1.5	2.2	3.5	4.1	
index_0	THC	2.7	2.95	2.99	3.24	2.43	2.39	2.69	3.02	2.59	
index_0	WD_HR	140	145	169	177	96	111	93	242	3	
index_0	WIND_D	120	115	173	155	104	173	74	303	289	
index_0	WIND_SI	0.4	0.5	0.4	0.4	0.5	0.6	0.7	0.5	1	
index_0	WS_HR	0.5	0.4	0.3	0.3	0.8	0.4	0.5	0.2	0.4	
index_1	AMB_TE	20.5	20.4	20.2	20	19.6	19.4	19.5	19.9	21.3	
index_1	CH4	2.33	2.37	2.66	2.56	2.32	2.27	2.39	2.5	2.45	
index_1	CO	0.68	0.64	0.69	0.63	0.4	0.36	0.5	0.79	0.92	

Kaggle Submission

- [Kaggle link](#)
- Display team name : <student ID>
- Submission format
 - A 245*2 .csv file, first row is for the column name and the last 244 rows for your result.
 - Column name must be **index** and **answer**.
 - [sample submission](#)
- There are one simple bassline and one strong bassline, beat them to get the higher score.

	A	B	C
1	index	answer	
2	index_0	0	
3	index_1	0	
4	index_2	0	
5	index_3	0	
6	index_4	0	
7	index_5	0	
8	index_6	0	

Kaggle Submission

- You can submit at most 5 times each day.
- The scoring metric is **RMSE**.
- You can choose 2 of the submissions to be considered for the private leaderboard, or will otherwise default to the best public scoring submissions.

Report Submission

Answer the following 3 questions:

1. How do you select features for your model input, what preprocessing did you do ?
2. Compare the impact of different training data amounts on the PM2.5 prediction accuracy, visualize the result and explain it.
3. Discuss the impact of regularization on PM2.5 prediction accuracy.

Please answer the question in detail !

Grading policy

- Kaggle (70%)
 - Basic score (according to the **private leaderboard**):
 - Over strong baseline : 55
 - Over simple bassline : 40
 - Under simple baseline : 25
 - Ranking score (according to the **private leaderboard**):
 $15 - (15/N) * (\text{ranking} - 1)$, N=numbers of people
- Source code and report (30%)
 - 10% for each quesiton

E3 Submission

Submit your source code and report to E3 before 4/11 23:59.

No late submission !

- Format

- source code : HW1_<student ID>.py or HW1_<student ID>.ipynb
- report : HW1_<student ID>.pdf

If you have any question about HW1, please feel free to contact with TA: WEI-LING HSU through email weiling.hsu.cs11@nycu.edu.tw